

# GENERAL CHEMISTRY

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## Chapter 16 Acid-Base Equilibria

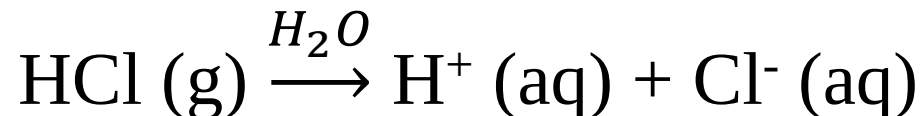
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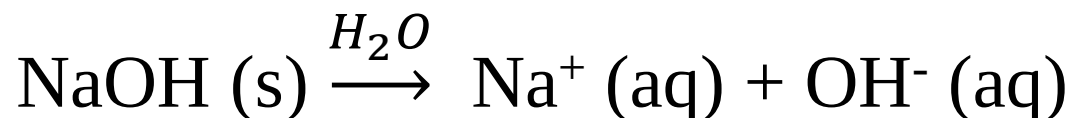
# 16-1 Acids and Bases: A Brief Review

## ◆ Arrhenius

- An **acid** is a substance that, when dissolved in water, increases the concentration of **hydrogen** ions, **H<sup>+</sup>**.



- A **base** is a substance that, when dissolved in water, increases the concentration of **hydroxide** ions, **OH<sup>-</sup>**.



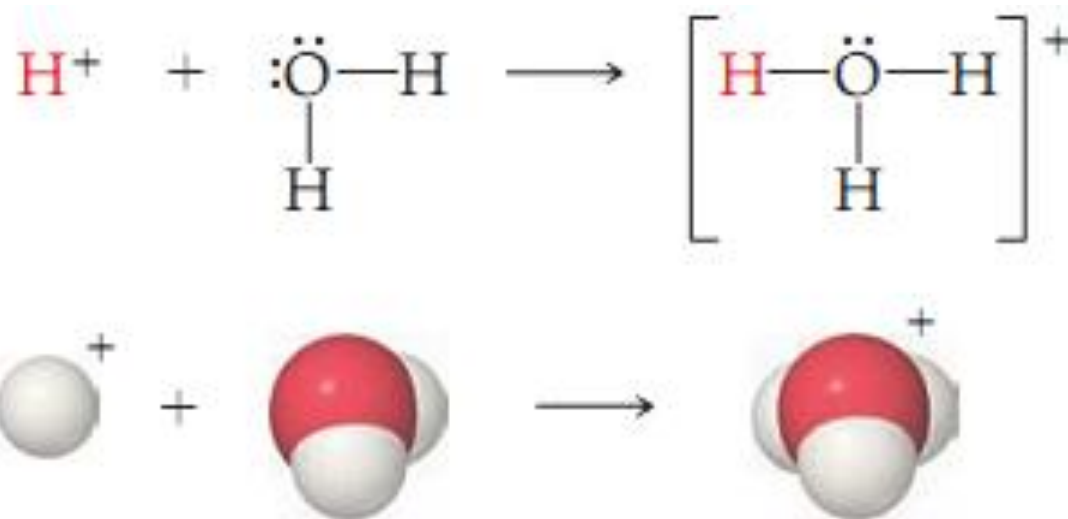
## 16-2 Brønsted–Lowry Acids and Bases

- ◆ **Brønsted–Lowry** : *Acid–base reactions involve the transfer of  $H^+$  ions from one substance to another*
  - An acid is a proton donor.
    - ... must have a removable proton
  - A base is a proton acceptor.
    - ...must have a pair of nonbonding electrons

# The H<sup>+</sup> Ion in Water



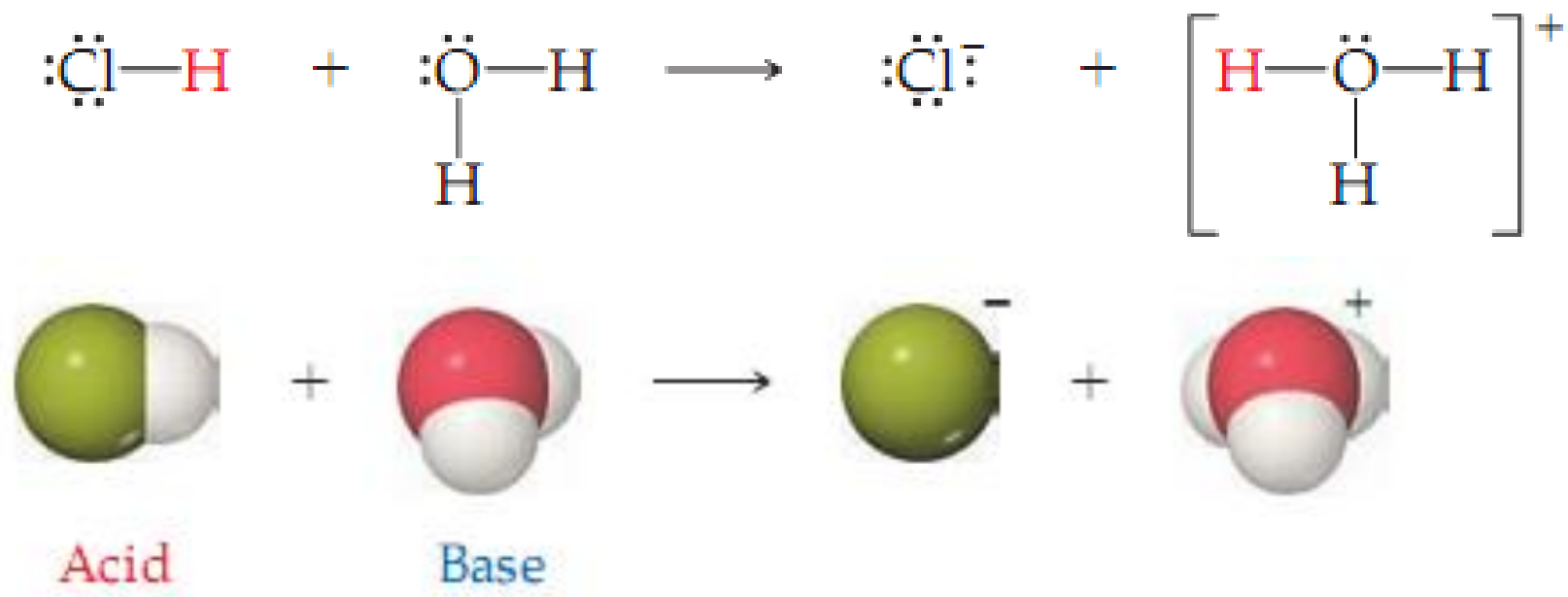
The H<sup>+</sup> ion interacts strongly with any source of electron density, such as the nonbonding electron pairs on the oxygen atoms of water molecules



Chemists use the notations H<sup>+</sup> and H<sub>3</sub>O<sup>+</sup> interchangeably

# Proton-Transfer Reactions

## What Happens When an Acid Dissolves in Water?

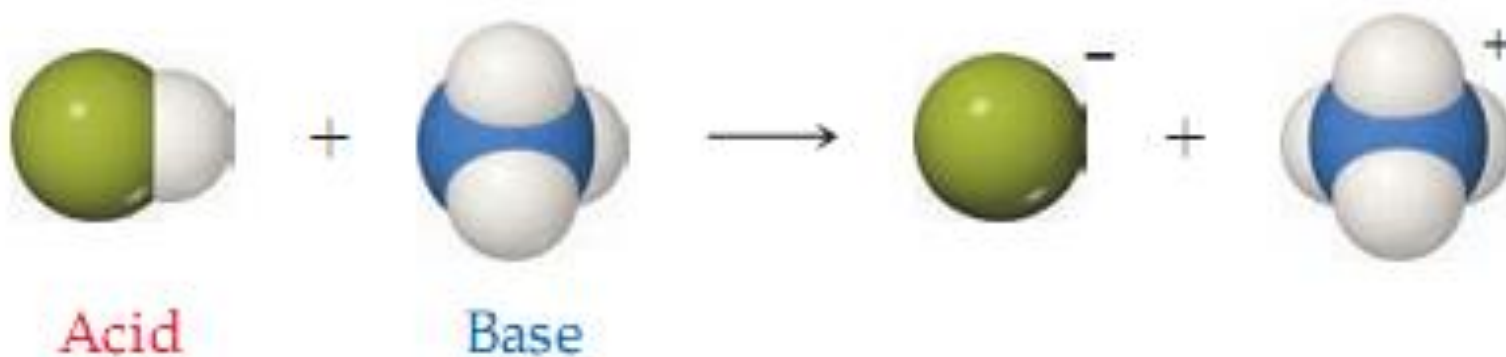
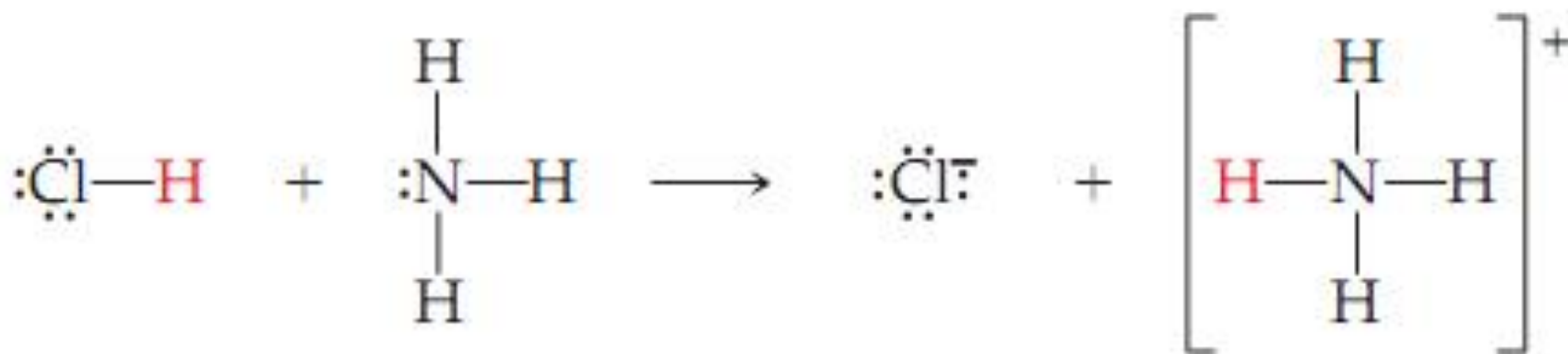


*proton donor    proton acceptor    conjugate base    hydronium ion*

- ◆ Water acts as a Brønsted–Lowry base and abstracts a proton from the acid.
- ◆ The **conjugate base** of the acid and a **hydronium ion** are formed.

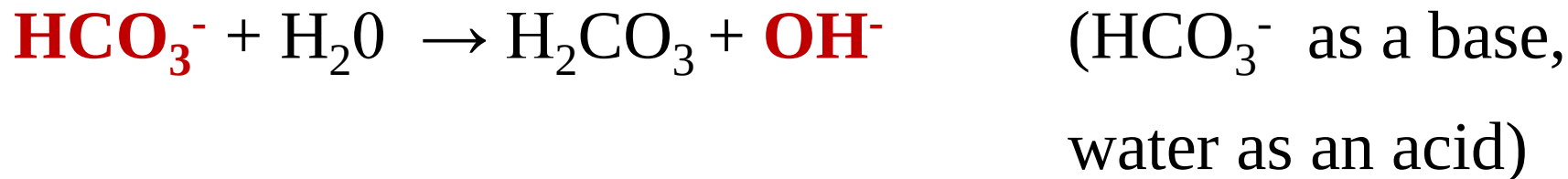
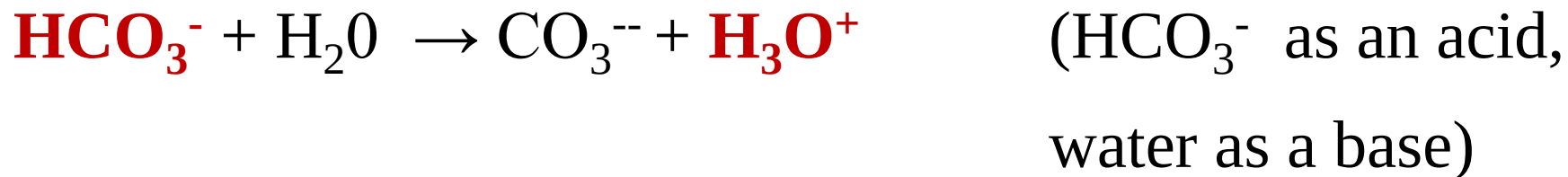
# Proton-Transfer Reactions

The Brønsted–Lowry concept also applies to reactions that do not occur in aqueous solutions



# Proton-Transfer Reactions

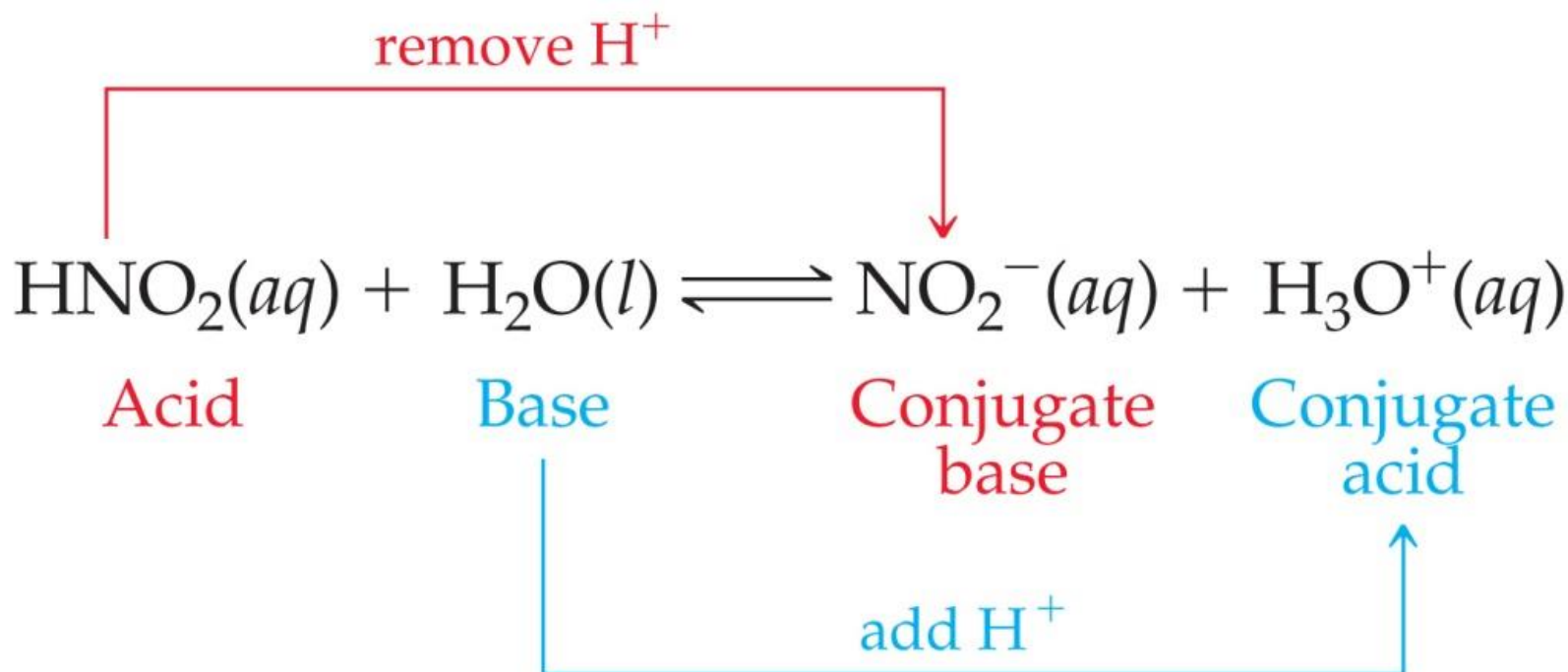
H<sub>2</sub>O and HCO<sub>3</sub><sup>-</sup> can be either acid or base



They are **amphiprotic**

# Conjugate Acids and Bases

- ◆ The term **conjugate** comes from the Latin word “conjugare,” meaning “to join together.”
- ◆ Reactions between acids and bases always yield their conjugate bases and acids.



# EXAMPLE

**(a) What is the conjugate base of**



**(b) What is the conjugate acid of**



# EXAMPLE

The hydrogen sulfite ion ( $\text{HSO}_3^-$ ) is amphoteric. Write an equation for the reaction of  $\text{HSO}_3^-$  with water

**(a)** in which the ion acts as an acid



acid

conjugate base

**(b)** in which the ion acts as a base.



base

conjugate acid

In both cases identify the conjugate acid–base pairs.

# Relative Strengths of Acids and Bases

| ACID               |                                             | BASE                                        |                                                                              |
|--------------------|---------------------------------------------|---------------------------------------------|------------------------------------------------------------------------------|
| Strong acids       | HCl                                         | Cl <sup>-</sup>                             | Negligible basicity                                                          |
|                    | H <sub>2</sub> SO <sub>4</sub>              | HSO <sub>4</sub> <sup>-</sup>               |                                                                              |
|                    | HNO <sub>3</sub>                            | NO <sub>3</sub> <sup>-</sup>                |                                                                              |
|                    | H <sub>3</sub> O <sup>+</sup> (aq)          | H <sub>2</sub> O                            |                                                                              |
|                    | HSO <sub>4</sub> <sup>-</sup>               | SO <sub>4</sub> <sup>2-</sup>               | Weak bases                                                                   |
|                    | H <sub>3</sub> PO <sub>4</sub>              | H <sub>2</sub> PO <sub>4</sub> <sup>-</sup> |                                                                              |
|                    | HF                                          | F <sup>-</sup>                              |                                                                              |
|                    | CH <sub>3</sub> COOH                        | CH <sub>3</sub> COO <sup>-</sup>            |                                                                              |
|                    | H <sub>2</sub> CO <sub>3</sub>              | HCO <sub>3</sub> <sup>-</sup>               |                                                                              |
|                    | H <sub>2</sub> S                            | HS <sup>-</sup>                             |                                                                              |
| Weak acids         | H <sub>2</sub> PO <sub>4</sub> <sup>-</sup> | HPO <sub>4</sub> <sup>2-</sup>              | Conjugate bases of substances with negligible acidity are exceedingly strong |
|                    | NH <sub>4</sub> <sup>+</sup>                | NH <sub>3</sub>                             |                                                                              |
|                    | HCO <sub>3</sub> <sup>-</sup>               | CO <sub>3</sub> <sup>2-</sup>               |                                                                              |
|                    | HPO <sub>4</sub> <sup>2-</sup>              | PO <sub>4</sub> <sup>3-</sup>               |                                                                              |
|                    | H <sub>2</sub> O                            | OH <sup>-</sup>                             |                                                                              |
| Negligible acidity | OH <sup>-</sup>                             | O <sup>2-</sup>                             | Strong bases                                                                 |
|                    | H <sub>2</sub>                              | H <sup>-</sup>                              |                                                                              |
|                    | CH <sub>4</sub>                             | CH <sub>3</sub> <sup>-</sup>                |                                                                              |

Strong acids are completely dissociated in water

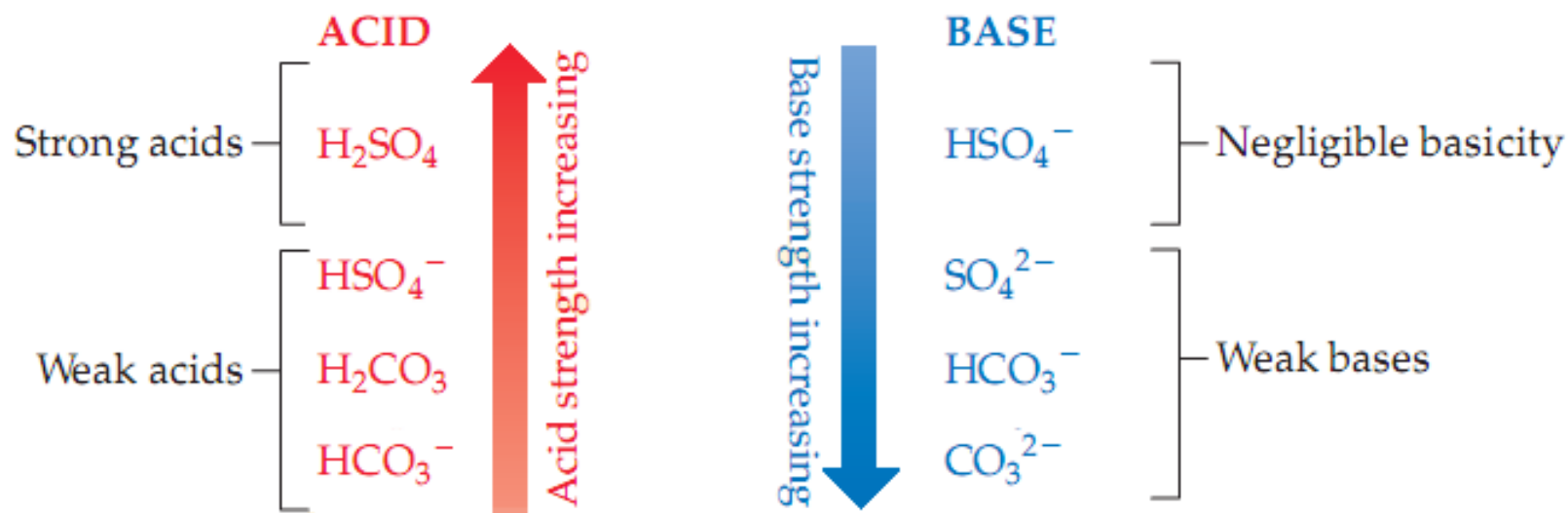
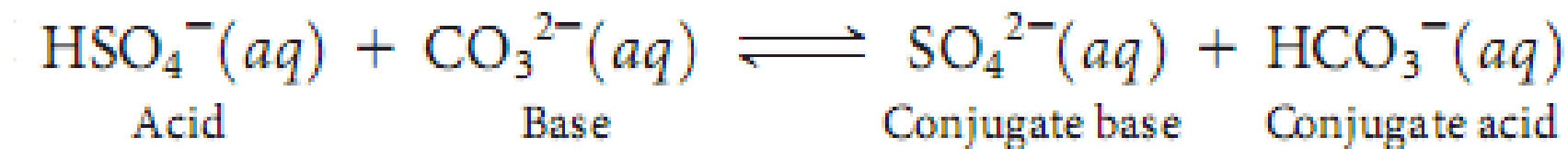
(The conjugate base of a strong acid shows negligible basicity)

Substances with negligible acidity do not dissociate in water

Conjugate bases of substances with negligible acidity are exceedingly strong

# EXAMPLE

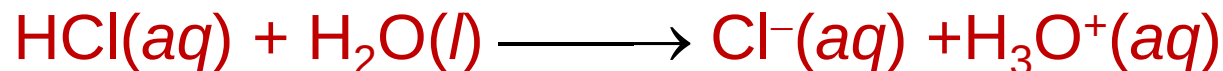
Predict whether the equilibrium of the following proton-transfer reaction lies to the left ( $K_c < 1$ ) or to the right ( $K_c > 1$ )?



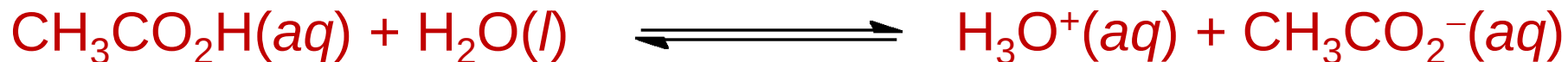
$\text{CO}_3^{2-}$  is the stronger base, therefore it will get the proton preferentially  
➔ The reaction lies to the right ( $K_c > 1$ )?

# Acid and Base Strength

- ◆ In any acid–base reaction, the equilibrium will favor the reaction that moves the proton to the stronger base to form the weaker acid and the weaker base.



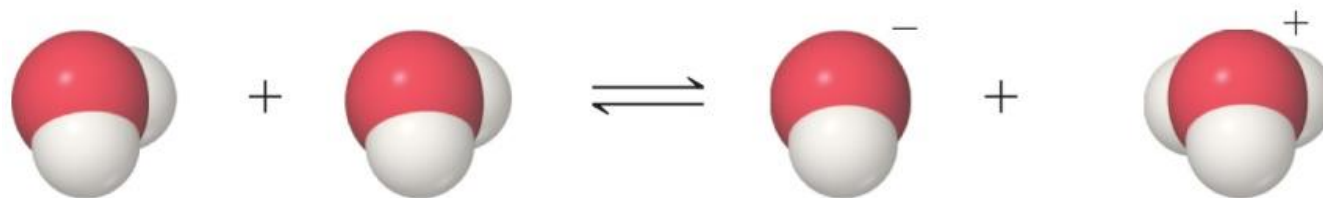
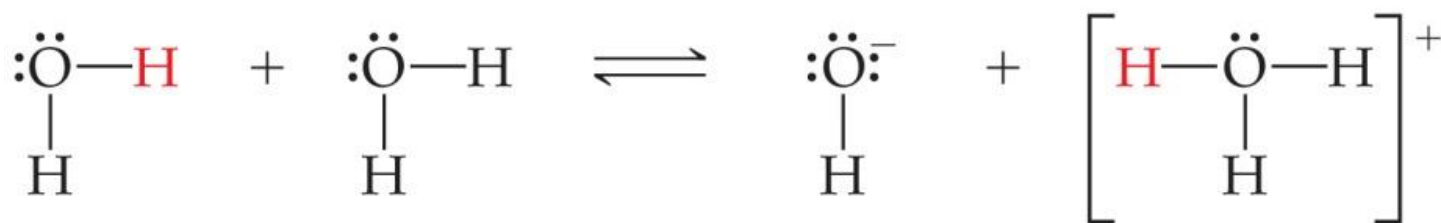
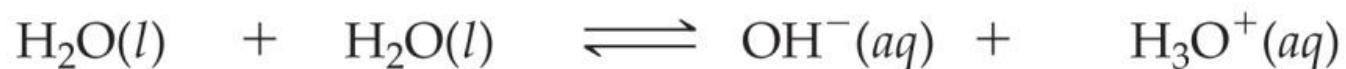
$\text{H}_2\text{O}$  is a much stronger base than  $\text{Cl}^-$ , so the equilibrium lies **completely to the right**



Acetate is a stronger base than  $\text{H}_2\text{O}$ , so **the reverse reaction is favored** more than the forward reaction

## 16-3 The Autoionization of Water

- ◆ As we have seen, water is amphoteric.
- ◆ In pure water, a few molecules act as bases and a few act as acids.
- ◆ This is referred to as **autoionization**.



Acid

Base

# The Ion Product of Water

- ◆ The equilibrium expression for the autoionization of water is

$$K_c = [\text{H}_3\text{O}^+] [\text{OH}^-]$$

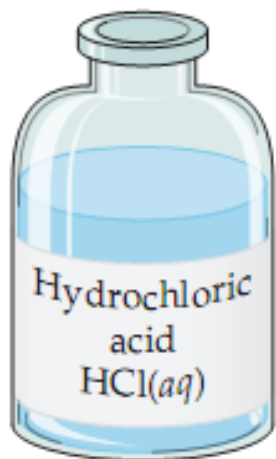
- ◆ This special equilibrium constant is referred to as the **ion product constant** for water,  $K_w$ .
- ◆ At 25°C,  $K_w = 1.0 \times 10^{-14}$

$$[\text{H}^+][\text{OH}^-] = (x)(x) = 1.0 \times 10^{-14}$$

$$x^2 = 1.0 \times 10^{-14}$$

$$x = 1.0 \times 10^{-7} \text{M} = [\text{H}^+] = [\text{OH}^-]$$

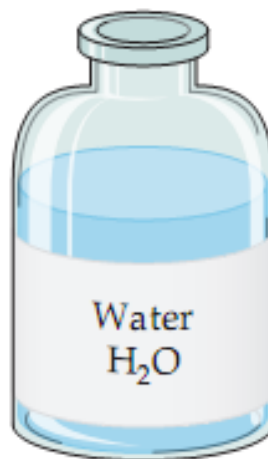
# The relative concentrations of $\text{H}^+$ and $\text{OH}^-$ in aqueous solutions at 25 °C



**Acidic solution**

$$[\text{H}^+] > [\text{OH}^-]$$

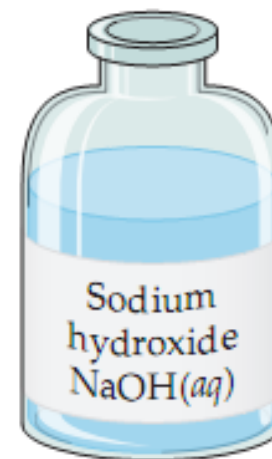
$$[\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$



**Neutral solution**

$$[\text{H}^+] = [\text{OH}^-]$$

$$[\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$



**Basic solution**

$$[\text{H}^+] < [\text{OH}^-]$$

$$[\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$

## EXAMPLE

Calculate the concentration of  $\text{H}^+(\text{aq})$  in

(a) a solution in which  $[\text{OH}^-]$  is 0.010 M,

(b) a solution in which  $[\text{OH}^-]$  is  $1.8 \times 10^{-9}$

The temperature of these solutions is  $25^\circ\text{C}$

$$[\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$

$$[\text{H}^+] = \frac{(1.0 \times 10^{-14})}{[\text{OH}^-]}$$

$$\text{a) } [\text{H}^+] = \frac{(1.0 \times 10^{-14})}{[\text{OH}^-]} = \frac{1.0 \times 10^{-14}}{0.010} = 1.0 \times 10^{-12} \text{ M}$$

$$\text{b) } [\text{H}^+] = \frac{(1.0 \times 10^{-14})}{[\text{OH}^-]} = \frac{1.0 \times 10^{-14}}{1.8 \times 10^{-9}} = 5.6 \times 10^{-6} \text{ M}$$

## 16-4 The pH scale

pH is the negative logarithm in base 10 of the concentration of hydronium ion:

$$\text{pH} = -\log [\text{H}_3\text{O}^+]$$

◆ **For pure water at 25°C,**

$$K_w = [\text{H}_3\text{O}^+] [\text{OH}^-] = 1.0 \times 10^{-14}$$

and

$$[\text{H}_3\text{O}^+] = [\text{OH}^-]$$

Thus,

$$[\text{H}_3\text{O}^+] = \sqrt{1.0 \times 10^{-14}} = 1.0 \times 10^{-7}$$

$$\text{pH} = -\log (1.0 \times 10^{-7}) = 7.00$$

# pH of Acids and Bases

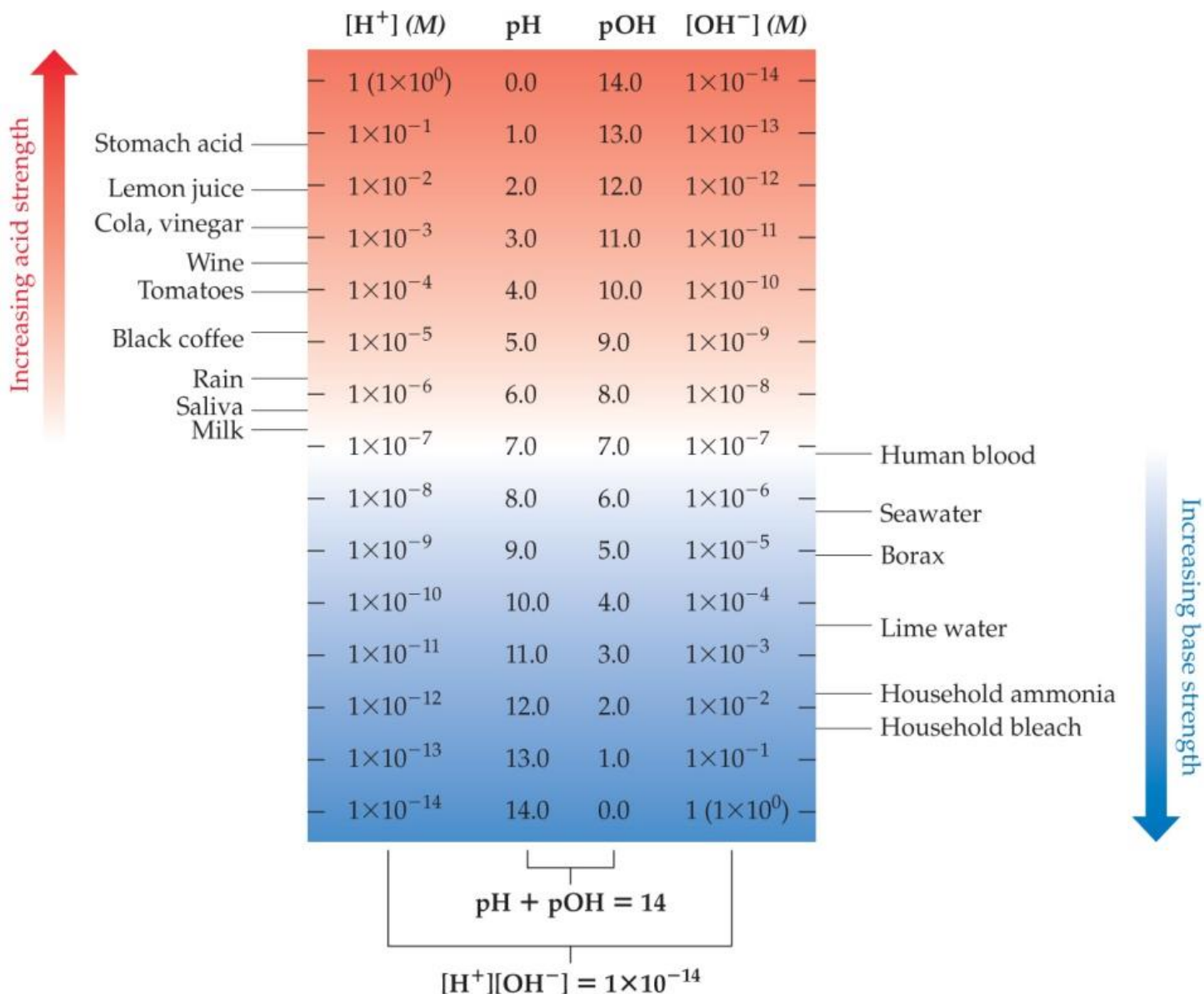
- ◆ An **acid** has a higher  $[\text{H}_3\text{O}^+]$  than pure water, so its **pH < 7**.
- ◆ A **base** has a lower  $[\text{H}_3\text{O}^+]$  than pure water, so its **pH > 7**.

TABLE 16.1 • Relationships among  $(\text{H}^+)$ ,  $(\text{OH}^-)$ , and pH at 25 °C

| Solution Type | $[\text{H}^+]$ (M)    | $[\text{OH}^-]$ (M)   | pH      |
|---------------|-----------------------|-----------------------|---------|
| Acidic        | $>1.0 \times 10^{-7}$ | $<1.0 \times 10^{-7}$ | $<7.00$ |
| Neutral       | $1.0 \times 10^{-7}$  | $1.0 \times 10^{-7}$  | 7.00    |
| Basic         | $<1.0 \times 10^{-7}$ | $>1.0 \times 10^{-7}$ | $>7.00$ |

# pH of Acids and Bases

These are the pH values for several common substances



# pOH and Other “p” Scales

- ◆ The “p” in pH tells us to take the negative logarithm in base 10 of a quantity (in this case, concentration of hydronium ions).
- ◆ Some similar examples are
  - pOH:  $-\log [\text{OH}^-]$
  - $\text{p}K_w$ :  $-\log K_w$

# pOH and Other “p” Scales

Because

$$[\text{H}_3\text{O}^+] [\text{OH}^-] = K_w = 1.0 \times 10^{-14}$$

we know that

$$-\log [\text{H}_3\text{O}^+] + -\log [\text{OH}^-] = -\log K_w = 14.00$$

or, in other words,

$$\text{pH} + \text{pOH} = \text{p}K_w = 14.00$$

## EXAMPLE

A sample of freshly pressed apple juice has a pOH of 10.24. Calculate  $[H^+]$ .

$$pH = 14.00 - pOH$$

$$pH = 14.00 - 10.24 = 3.76$$

$$pH = -\log[H^+] = 3.76$$

$$\log[H^+] = -3.76$$

$$[H^+] = \text{antilog}(-3.76) = 10^{-3.76} = 1.7 \times 10^{-4} M$$

# Measuring pH

- ◆ For less accurate measurements, one can use litmus paper or an indicator

|                   | pH range for color change |                                                                                   |                                                                                    |                                                                                     |                                                                                       |                                                                                       |                                                                                       |     |  |  |
|-------------------|---------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----|--|--|
|                   | 0                         | 2                                                                                 | 4                                                                                  | 6                                                                                   | 8                                                                                     | 10                                                                                    | 12                                                                                    | 14  |  |  |
| Methyl violet     | Yellow                    |  | Violet                                                                             |                                                                                     |                                                                                       |                                                                                       |                                                                                       |     |  |  |
| Thymol blue       | Red                       |  | Yellow                                                                             |                                                                                     | Yellow                                                                                |    | Blue                                                                                  |     |  |  |
| Methyl orange     |                           | Red                                                                               |  | Yellow                                                                              |                                                                                       |                                                                                       |                                                                                       |     |  |  |
| Methyl red        |                           |                                                                                   | Red                                                                                |  | Yellow                                                                                |                                                                                       |                                                                                       |     |  |  |
| Bromthymol blue   |                           |                                                                                   |                                                                                    | Yellow                                                                              |  | Blue                                                                                  |                                                                                       |     |  |  |
| Phenolphthalein   |                           |                                                                                   |                                                                                    |                                                                                     | Colorless                                                                             |  | Pink                                                                                  |     |  |  |
| Alizarin yellow R |                           |                                                                                   |                                                                                    |                                                                                     |                                                                                       | Yellow                                                                                |  | Red |  |  |

# Measuring pH

For more accurate measurements, one uses a pH meter, which measures the voltage in the solution.



# 16.5 Strong Acids and Bases

|                    | ACID                                        |                                                                                                                | BASE                                        |                     |
|--------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------|
| Strong acids       | HCl                                         | <div> <div>↑</div> <div>Base strength increasing</div> <div>↓</div> <div>Acid strength increasing</div> </div> | Cl <sup>-</sup>                             | Negligible basicity |
|                    | H <sub>2</sub> SO <sub>4</sub>              |                                                                                                                | HSO <sub>4</sub> <sup>-</sup>               |                     |
|                    | HNO <sub>3</sub>                            |                                                                                                                | NO <sub>3</sub> <sup>-</sup>                |                     |
|                    | H <sub>3</sub> O <sup>+</sup> (aq)          |                                                                                                                | H <sub>2</sub> O                            |                     |
| Weak acids         | HSO <sub>4</sub> <sup>-</sup>               |                                                                                                                | SO <sub>4</sub> <sup>2-</sup>               | Weak bases          |
|                    | H <sub>3</sub> PO <sub>4</sub>              |                                                                                                                | H <sub>2</sub> PO <sub>4</sub> <sup>-</sup> |                     |
|                    | HF                                          |                                                                                                                | F <sup>-</sup>                              |                     |
|                    | CH <sub>3</sub> COOH                        |                                                                                                                | CH <sub>3</sub> COO <sup>-</sup>            |                     |
|                    | H <sub>2</sub> CO <sub>3</sub>              |                                                                                                                | HCO <sub>3</sub> <sup>-</sup>               |                     |
|                    | H <sub>2</sub> S                            |                                                                                                                | HS <sup>-</sup>                             |                     |
|                    | H <sub>2</sub> PO <sub>4</sub> <sup>-</sup> |                                                                                                                | HPO <sub>4</sub> <sup>2-</sup>              |                     |
|                    | NH <sub>4</sub> <sup>+</sup>                |                                                                                                                | NH <sub>3</sub>                             |                     |
|                    | HCO <sub>3</sub> <sup>-</sup>               |                                                                                                                | CO <sub>3</sub> <sup>2-</sup>               |                     |
|                    | HPO <sub>4</sub> <sup>2-</sup>              |                                                                                                                | PO <sub>4</sub> <sup>3-</sup>               |                     |
| Negligible acidity | H <sub>2</sub> O                            |                                                                                                                | OH <sup>-</sup>                             | Strong bases        |
|                    | OH <sup>-</sup>                             |                                                                                                                | O <sup>2-</sup>                             |                     |
|                    | H <sub>2</sub>                              |                                                                                                                | H <sup>-</sup>                              |                     |
|                    | CH <sub>4</sub>                             |                                                                                                                | CH <sub>3</sub> <sup>-</sup>                |                     |

# Strong Acids

- ◆ The seven most common strong acids include
  - six monoprotic acids HCl, HBr, HI, HNO<sub>3</sub>, HClO<sub>3</sub>, HClO<sub>4</sub> and
  - one diprotic acid H<sub>2</sub>SO<sub>4</sub>.
- ◆ They are strong electrolytes and exist totally as ions in aqueous solution.
- ◆ For the **monoprotic strong acids**,  
 **$[\text{H}_3\text{O}^+] = [\text{acid}]$**

Example: In a 0.20 M solution of HNO<sub>3</sub>(aq),  $[\text{H}^+] = [\text{NO}_3^-] = 0.20 \text{ M}$

# Strong Bases

- ◆ Strong bases are the soluble hydroxides, which are the alkali metal and heavier alkaline earth metal hydroxides ( $\text{Ca}^{2+}$ ,  $\text{Sr}^{2+}$ , and  $\text{Ba}^{2+}$ ).
- ◆ These substances dissociate completely in aqueous solution.

## EXAMPLE

What is the pH of

(a) a 0.040 M solution of  $\text{HClO}_4$

Because  $\text{HClO}_4$  is a strong acid, it is completely ionized in water, giving  $[\text{H}^+] = [\text{ClO}_4^-] = 0.040 \text{ M}$

$$\text{pH} = -\log(0.040) = 1.40$$

(b) a 0.0011 M solution of  $\text{Ca}(\text{OH}_2)$

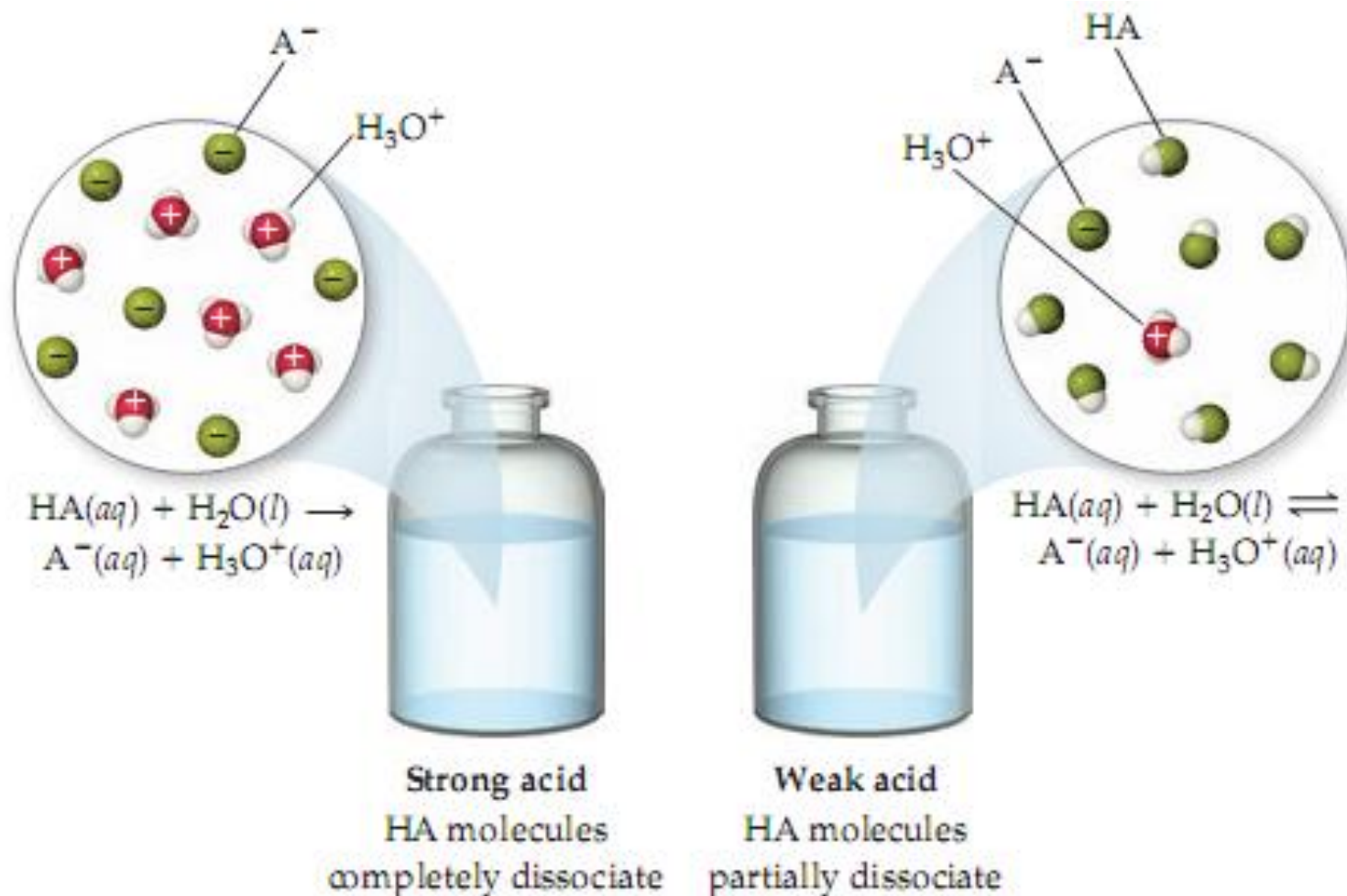
$\text{Ca}(\text{OH}_2)$  is a strong base that dissociates in water to give two  $\text{OH}^-$  ions per formula unit, giving  $[\text{OH}^-] = 2 \times 0.0011 = 0.0022 \text{ M}$

$$\text{pOH} = -\log(0.0022) = 2.66$$

$$\text{pH} = 14.00 - \text{pOH} = 11.34$$

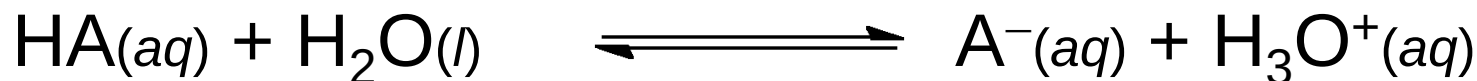
## 16-6 Weak Acids

- ◆ Most acidic substances are weak acids and therefore only partially ionized in aqueous solution



# Dissociation Constants

- ◆ For a generalized acid dissociation,



the equilibrium expression would be

$$K_c = \frac{[\text{H}_3\text{O}^{+}] [\text{A}^{-}]}{[\text{HA}]}$$

- ◆ This equilibrium constant is called the **acid-dissociation constant**,  $K_a$ .

# Dissociation Constants

The greater the value of  $K_a$ , the stronger is the acid.

TABLE 16.2 • Some Weak Acids in Water at 25 °C

| Acid                                          | Structural Formula*                                          | Conjugate Base                     | $K_a$                 |
|-----------------------------------------------|--------------------------------------------------------------|------------------------------------|-----------------------|
| Chlorous ( $\text{HClO}_2$ )                  | $\text{H}-\text{O}-\text{Cl}-\text{O}$                       | $\text{ClO}_2^-$                   | $1.0 \times 10^{-2}$  |
| Hydrofluoric ( $\text{HF}$ )                  | $\text{H}-\text{F}$                                          | $\text{F}^-$                       | $6.8 \times 10^{-4}$  |
| Nitrous ( $\text{HNO}_2$ )                    | $\text{H}-\text{O}-\text{N}=\text{O}$                        | $\text{NO}_2^-$                    | $4.5 \times 10^{-4}$  |
| Benzoic ( $\text{C}_6\text{H}_5\text{COOH}$ ) | $\text{H}-\text{O}-\text{C}(=\text{O})-\text{C}_6\text{H}_5$ | $\text{C}_6\text{H}_5\text{COO}^-$ | $6.3 \times 10^{-5}$  |
| Acetic ( $\text{CH}_3\text{COOH}$ )           | $\text{H}-\text{O}-\text{C}(=\text{O})-\text{CH}_3$          | $\text{CH}_3\text{COO}^-$          | $1.8 \times 10^{-5}$  |
| Hypochlorous ( $\text{HOCl}$ )                | $\text{H}-\text{O}-\text{Cl}$                                | $\text{OCl}^-$                     | $3.0 \times 10^{-5}$  |
| Hydrocyanic ( $\text{HCN}$ )                  | $\text{H}-\text{C}\equiv\text{N}$                            | $\text{CN}^-$                      | $4.9 \times 10^{-10}$ |
| Phenol ( $\text{HOC}_6\text{H}_5$ )           | $\text{H}-\text{O}-\text{C}_6\text{H}_5$                     | $\text{C}_6\text{H}_5\text{O}^-$   | $1.3 \times 10^{-10}$ |

\*The proton that ionizes is shown in red.

# Calculating $K_a$ from the pH

The pH of a 0.10 M solution of formic acid, HCOOH, at 25°C is 2.38. Calculate  $K_a$  for formic acid at this temperature.

We know that

$$K_a = \frac{[\text{H}^+][\text{HCOO}^-]}{[\text{HCOOH}]}$$

$$\text{pH} = -\log[\text{H}^+] = 2.38$$

$$\log[\text{H}^+] = -2.38$$

$$[\text{H}^+] = 10^{-2.38} = 4.2 \times 10^{-3} \text{ M}$$

The dissociation equation:  $\text{HCOOH}(aq) \rightleftharpoons \text{H}^+(aq) + \text{HCOO}^-(aq)$

Using the ICE chart

|   |      |   |   |
|---|------|---|---|
| I | 0.10 | 0 | 0 |
| C |      |   |   |
| E |      |   |   |

$$K_a = \frac{(4.2 \times 10^{-3})(4.2 \times 10^{-3})}{(0.10 - 4.2 \times 10^{-3})} = 1.8 \times 10^{-4}$$

# Percent Ionization

◆ Percent ionization =  $\frac{[\text{H}_3\text{O}^+]_{\text{eq}}}{[\text{HA}]_{\text{initial}}} \times 100$

◆ In this example,

|   |                             |                       |                       |
|---|-----------------------------|-----------------------|-----------------------|
| I | 0.10                        | 0                     | 0                     |
| C | $-4.2 \times 10^{-3}$       | $+4.2 \times 10^{-3}$ | $+4.2 \times 10^{-3}$ |
| E | $0.10 - 4.2 \times 10^{-3}$ | $4.2 \times 10^{-3}$  | $4.2 \times 10^{-3}$  |

$$[\text{H}_3\text{O}^+]_{\text{eq}} = 4.2 \times 10^{-3} \text{ M}$$

$$[\text{HCOOH}]_{\text{initial}} = 0.10 \text{ M}$$

$$\text{Percent ionization} = \frac{4.2 \times 10^{-3}}{0.10} \times 100 = 4.2\%$$

# Calculating pH from $K_a$

1. Write the ionization equilibrium
2. Write the equilibrium-constant expression and the value for the equilibrium constant
3. Express the concentrations involved in the equilibrium reaction using ICE chart
4. Substitute the equilibrium concentrations into the equilibrium-constant expression and solve for change in concentration
5. Calculate pH

# Calculating pH from $K_a$

Calculate the pH of a 0.30 M solution of acetic acid,  $\text{HC}_2\text{H}_3\text{O}_2$ , at  $25^\circ\text{C}$ .  $K_a$  for acetic acid at  $25^\circ\text{C}$  is  $1.8 \times 10^{-5}$ .



2.  $K_a = \frac{[\text{H}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = 1.8 \times 10^{-5}$



|   |        |    |    |
|---|--------|----|----|
| I | 0.30   | 0  | 0  |
| C | -x     | +x | +x |
| E | 0.30-x | x  | x  |

4.  $K_a = \frac{[\text{H}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = \frac{(x)(x)}{0.30 - x} = 1.8 \times 10^{-5}$

$$x = 2.3 \times 10^{-3} \text{ M} = [\text{H}^+]$$

5.  $\text{pH} = -\log(2.3 \times 10^{-3}) = 2.64$

# EXAMPLE

Calculate the pH of a 0.20 M solution of HCN.  $K_a$  for HCN is  $4.9 \times 10^{-10}$



$$K_a = \frac{[\text{H}^+][\text{CN}^-]}{[\text{HCN}]} = 4.9 \times 10^{-10}$$



|   |        |    |    |
|---|--------|----|----|
| I | 0.20   | 0  | 0  |
| C | -x     | +x | +x |
| E | 0.20-x | x  | x  |

$$K_a = \frac{(x)(x)}{0.20 - x} = 4.9 \times 10^{-10}$$

$$x = \sqrt{0.98 \times 10^{-10}} = 9.9 \times 10^{-6} \text{ M} = [\text{H}^+]$$

$$\text{pH} = -\log[\text{H}^+] = -\log(9.9 \times 10^{-6}) = 5.00$$

# Polyprotic Acids

**Polyprotic acids have more than one acidic proton.**



- ◆ It is always easier to remove the first proton from a polyprotic acid than to remove the second. Therefore  $K_{a1} > K_{a2}$
- ◆ For an acid with three ionizable protons, it is easier to remove the second proton than the third. Therefore,  $K_{a2} > K_{a3}$
- ◆ If the difference between the  $K_a$  for the first dissociation and subsequent  $K_a$  values is  $10^3$  or more, the pH generally depends *only* on the first dissociation.

# Polyprotic Acids

**TABLE 16.3 • Acid-Dissociation Constants of Some Common Polyprotic Acids**

| Name       | Formula                                    | $K_{a1}$             | $K_{a2}$              | $K_{a3}$              |
|------------|--------------------------------------------|----------------------|-----------------------|-----------------------|
| Ascorbic   | $\text{H}_2\text{C}_6\text{H}_6\text{O}_6$ | $8.0 \times 10^{-5}$ | $1.6 \times 10^{-12}$ |                       |
| Carbonic   | $\text{H}_2\text{CO}_3$                    | $4.3 \times 10^{-7}$ | $5.6 \times 10^{-11}$ |                       |
| Citric     | $\text{H}_3\text{C}_6\text{H}_5\text{O}_7$ | $7.4 \times 10^{-4}$ | $1.7 \times 10^{-5}$  | $4.0 \times 10^{-7}$  |
| Oxalic     | $\text{H}_2\text{C}_2\text{O}_4$           | $5.9 \times 10^{-2}$ | $6.4 \times 10^{-5}$  |                       |
| Phosphoric | $\text{H}_3\text{PO}_4$                    | $7.5 \times 10^{-3}$ | $6.2 \times 10^{-8}$  | $4.2 \times 10^{-13}$ |
| Sulfurous  | $\text{H}_2\text{SO}_3$                    | $1.7 \times 10^{-2}$ | $6.4 \times 10^{-8}$  |                       |
| Sulfuric   | $\text{H}_2\text{SO}_4$                    | Large                | $1.2 \times 10^{-2}$  |                       |
| Tartaric   | $\text{H}_2\text{C}_4\text{H}_4\text{O}_6$ | $1.0 \times 10^{-3}$ | $4.6 \times 10^{-5}$  |                       |

# EXAMPLE

The solubility of  $\text{CO}_2$  in water at  $25^\circ\text{C}$  and  $0.1\text{ atm}$  is  $0.0037\text{ M}$ . The common practice is to assume that all the dissolved  $\text{CO}_2$  is in the form of carbonic acid ( $\text{H}_2\text{CO}_3$ ), which is produced in the reaction:



(a) What is the pH of a  $0.0037\text{ M}$  solution of  $\text{H}_2\text{CO}_3$ ?

(b) What is the concentration of  $\text{CO}_3^{2-}$ ?

For  $\text{H}_2\text{CO}_3$ ,  $K_{a1} = 4.3 \times 10^{-7}$ ,  $K_{a2} = 5.6 \times 10^{-11}$

$K_{a1}$  and  $K_{a2}$  differ by more than a factor of  $10^3$  the pH can be determined by considering only  $K_{a1}$



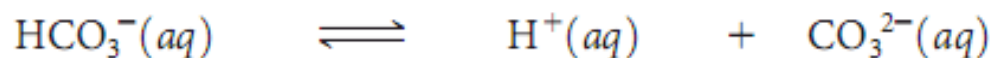
|   |          |    |    |
|---|----------|----|----|
| I | 0.0037   | 0  | 0  |
| C | -x       | +x | +x |
| E | 0.0037-x | x  | x  |

$$K_{a1} = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]} = \frac{(x)(x)}{0.0037 - x} = 4.3 \times 10^{-7}$$

$$x = 4.0 \times 10^{-5}\text{ M}$$

$$\text{pH} = -\log(4.0 \times 10^{-5}) = 4.40$$

# EXAMPLE



|   |                          |                      |    |
|---|--------------------------|----------------------|----|
| I | $4.0 \times 10^{-5}$     | $4.0 \times 10^{-5}$ | 0  |
| C | -y                       | +y                   | +y |
| E | $4.0 \times 10^{-5} - y$ | y                    | y  |

$$K_{a2} = \frac{[\text{H}^+][\text{CO}_3^{2-}]}{[\text{HCO}_3^-]} = \frac{(4.0 \times 10^{-5})(y)}{4.0 \times 10^{-5}} = 5.6 \times 10^{-11}$$

$$y = 5.6 \times 10^{-11} \text{ M} = [\text{CO}_3^{2-}]$$

# 16-7 Weak Bases

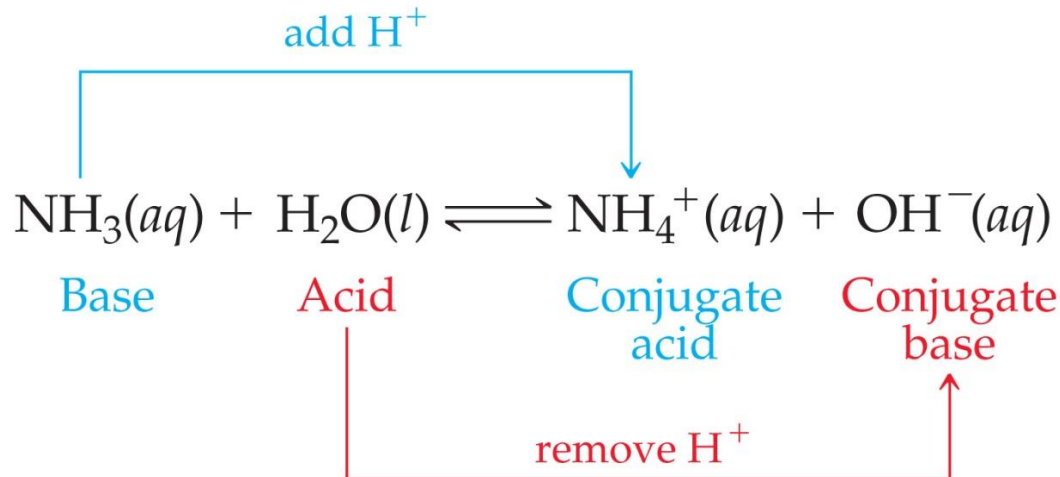
Bases react with water to produce hydroxide ion.



The equilibrium constant expression for this reaction is

$$K_b = \frac{[BH^+][OH^-]}{[B]}$$

$K_b$  is the base-dissociation constant



$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]}$$

$K_b$  can be used to find  $[\text{OH}^-]$  and, through it, pH.

**TABLE 16.4 • Some Weak Bases in Water at 25 °C**

| Base                                        | Structural Formula*                                                                                                                                   | Conjugate Acid                    | $K_b$                |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----------------------|
| Ammonia ( $\text{NH}_3$ )                   | $\begin{array}{c} \text{H}-\ddot{\text{N}}-\text{H} \\   \\ \text{H} \end{array}$                                                                     | $\text{NH}_4^+$                   | $1.8 \times 10^{-5}$ |
| Pyridine ( $\text{C}_5\text{H}_5\text{N}$ ) |                                                                      | $\text{C}_5\text{H}_5\text{NH}^+$ | $1.7 \times 10^{-9}$ |
| Hydroxylamine ( $\text{HONH}_2$ )           | $\begin{array}{c} \text{H}-\ddot{\text{N}}-\ddot{\text{O}}\text{H} \\   \\ \text{H} \end{array}$                                                      | $\text{HONH}_3^+$                 | $1.1 \times 10^{-8}$ |
| Methylamine ( $\text{CH}_3\text{NH}_2$ )    | $\begin{array}{c} \text{H}-\ddot{\text{N}}-\text{CH}_3 \\   \\ \text{H} \end{array}$                                                                  | $\text{CH}_3\text{NH}_3^+$        | $4.4 \times 10^{-4}$ |
| Hydrosulfide ion ( $\text{HS}^-$ )          | $\left[ \text{H}-\ddot{\text{S}} \right]^-$                                                                                                           | $\text{H}_2\text{S}$              | $1.8 \times 10^{-7}$ |
| Carbonate ion ( $\text{CO}_3^{2-}$ )        | $\left[ \begin{array}{c} \ddot{\text{O}}: \\   \\ \text{C} \\ / \quad \backslash \\ \ddot{\text{O}}: \quad \ddot{\text{O}}: \end{array} \right]^{2-}$ | $\text{HCO}_3^-$                  | $1.8 \times 10^{-4}$ |
| Hypochlorite ion ( $\text{ClO}^-$ )         | $\left[ :\ddot{\text{Cl}}-\ddot{\text{O}}: \right]^-$                                                                                                 | $\text{HClO}$                     | $3.3 \times 10^{-7}$ |

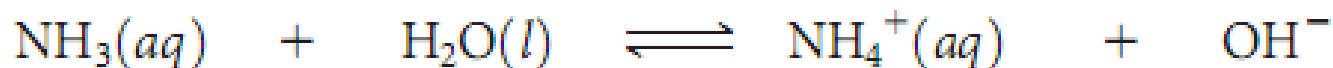
\*The atom that accepts the proton is shown in blue.

# EXAMPLE

What is the pH of a 0.15 M solution of  $\text{NH}_3$ ? For  $\text{NH}_3$ ,  $K_b = 1.8 \times 10^{-5}$



$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]} = 1.8 \times 10^{-5}$$



|   |        |  |    |    |
|---|--------|--|----|----|
| I | 0.15   |  | 0  | 0  |
| C | -x     |  | +x | +x |
| E | 0.15-x |  | x  | x  |

$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]} = \frac{(x)(x)}{0.15 - x} = 1.8 \times 10^{-5}$$

$$x = [\text{NH}_4^+] = [\text{OH}^-] = \sqrt{2.7 \times 10^{-6}} = 1.6 \times 10^{-3} M$$

$$\text{pOH} = -\log(1.6 \times 10^{-3}) = 2.80$$

$$\text{pH} = 14.00 - 2.80 = 11.20$$

## 16-8 Relationship Between $K_a$ and $K_b$

The product of the acid-dissociation constant for an acid and the base-dissociation constant for its conjugate base equals the ion-product constant for water:

$$K_a \times K_b = K_w$$
$$pK_a + pK_b = pK_w = 14 \text{ at } 25^\circ\text{C}$$



$$K_a = \frac{[\text{NH}_3][\text{H}^+]}{[\text{NH}_4^+]}$$



$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]}$$



$$K_a \times K_b = \left( \frac{[\text{NH}_3][\text{H}^+]}{[\text{NH}_4^+]} \right) \left( \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]} \right)$$
$$= [\text{H}^+][\text{OH}^-] = K_w$$

**Table 16.5 Some Conjugate Acid–Base Pairs**

| Acid                           | $K_a$                 | Base                             | $K_b$                 |
|--------------------------------|-----------------------|----------------------------------|-----------------------|
| HNO <sub>3</sub>               | (Strong acid)         | NO <sub>3</sub> <sup>−</sup>     | (Negligible basicity) |
| HF                             | $6.8 \times 10^{-4}$  | F <sup>−</sup>                   | $1.5 \times 10^{-11}$ |
| CH <sub>3</sub> COOH           | $1.8 \times 10^{-5}$  | CH <sub>3</sub> COO <sup>−</sup> | $5.6 \times 10^{-10}$ |
| H <sub>2</sub> CO <sub>3</sub> | $4.3 \times 10^{-7}$  | HCO <sub>3</sub> <sup>−</sup>    | $2.3 \times 10^{-8}$  |
| NH <sub>4</sub> <sup>+</sup>   | $5.6 \times 10^{-10}$ | NH <sub>3</sub>                  | $1.8 \times 10^{-5}$  |
| HCO <sub>3</sub> <sup>−</sup>  | $5.6 \times 10^{-11}$ | CO <sub>3</sub> <sup>2−</sup>    | $1.8 \times 10^{-4}$  |
| OH <sup>−</sup>                | (Negligible acidity)  | O <sup>2−</sup>                  | (Strong base)         |

# 16-9 Acid–Base Properties of Salt Solutions

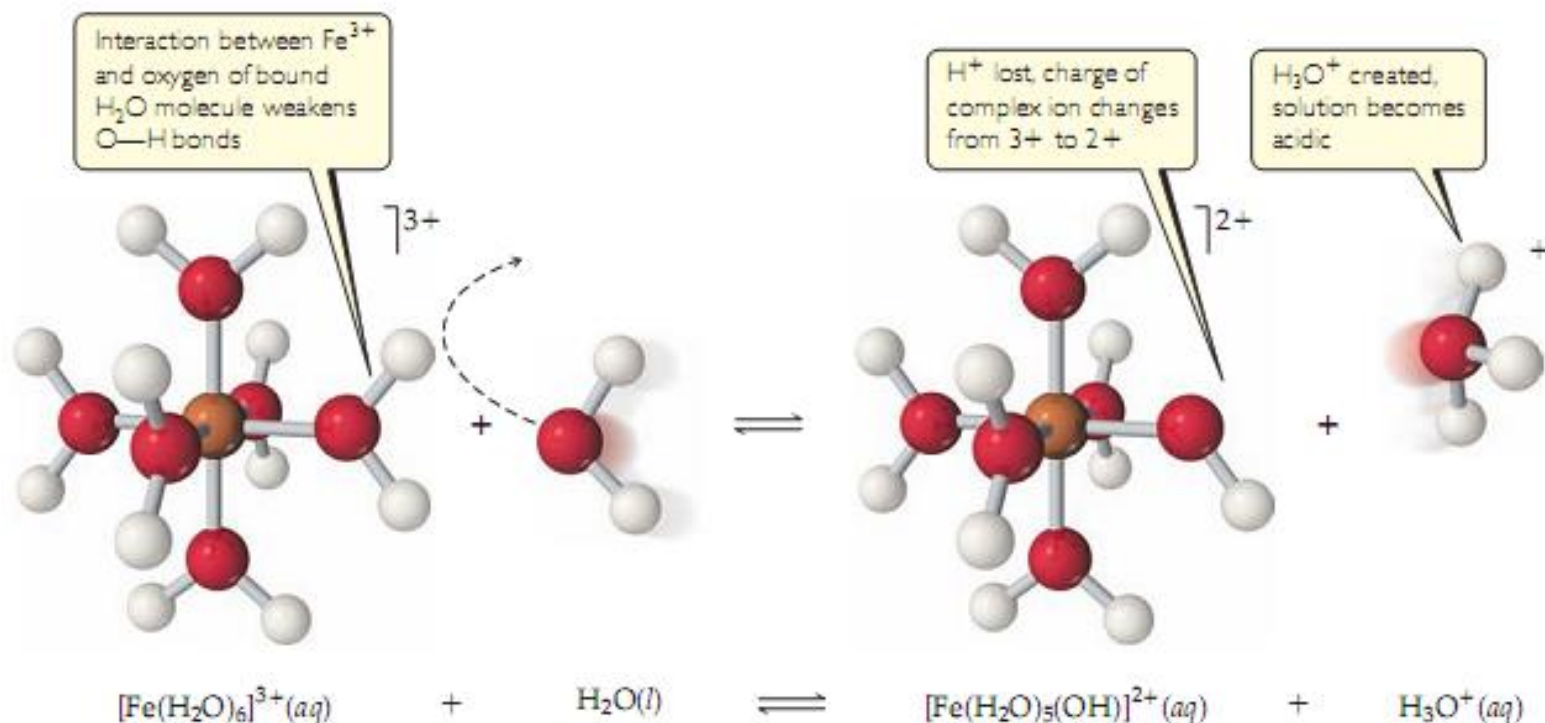
- ◆ Anions can be considered the conjugate bases of acids.
  - As such, they can react with water in a **hydrolysis reaction** to form  $\text{OH}^-$  and the conjugate acid:



- ◆ Polyatomic cations containing one or more protons can be considered the conjugate acids of weak bases.
  - As such, they can **donate a proton to water**, producing hydronium ions and thereby lowering the pH



# Reactions of Cations with Water



- ◆ Attraction between nonbonding electrons on oxygen and the metal causes a shift of the electron density in water.
- ◆ This makes the O—H bond more polar and the water more acidic.

# Reactions of Cations with Water

**TABLE 16.6 • Acid-Dissociation Constants for Metal Cations in Aqueous Solution at 25 °C**

| Cation           | $K_a$                 |
|------------------|-----------------------|
| $\text{Fe}^{2+}$ | $3.2 \times 10^{-10}$ |
| $\text{Zn}^{2+}$ | $2.5 \times 10^{-10}$ |
| $\text{Ni}^{2+}$ | $2.5 \times 10^{-11}$ |
| $\text{Fe}^{3+}$ | $6.3 \times 10^{-3}$  |
| $\text{Cr}^{3+}$ | $1.6 \times 10^{-4}$  |
| $\text{Al}^{3+}$ | $1.4 \times 10^{-5}$  |

- ◆ Most metal cations that are hydrated in solution also lower the pH of the solution
- ◆ Attraction between nonbonding electrons on oxygen and the metal causes a shift of the electron density in water.
- ◆ This makes the O–H bond more polar and the water more acidic.
- ◆ Greater charge and smaller size make a cation more acidic

# Combined Effect of Cation and Anion in Solution

1. An anion that is the conjugate base of a strong acid will not affect the pH.
2. An anion that is the conjugate base of a weak acid will increase the pH.
3. A cation that is the conjugate acid of a weak base will decrease the pH.
4. Cations of the strong Arrhenius bases will not affect the pH.
5. Other metal ions will cause a decrease in pH.
6. When a solution contains both a weak acid and a weak base, the affect on pH depends on the  $K_a$  and  $K_b$  values.

# EXAMPLE

Determine whether aqueous solutions of each of these salts are acidic, basic, or neutral:

(a)  $\text{Ba}(\text{CH}_3\text{COO})_2$       **Basic**

(b)  $\text{NH}_4\text{Cl}$       **Acidic**

(c)  $\text{CH}_3\text{NH}_3\text{Br}$ ,      **Acidic**

(d)  $\text{KNO}_3$ ,      **Neutral**

(e)  $\text{Al}(\text{ClO}_4)_3$       **Acidic**

# 16-10 Acid–Base Behavior and Chemical Structure

## Factors Affecting Acid Strength

- ◆ The more polar the H–X bond the more acidic the compound
- ◆ The weaker the H–X bond, the more acidic the compound.
- ◆ The greater the stability of the conjugate base, the stronger the acid.

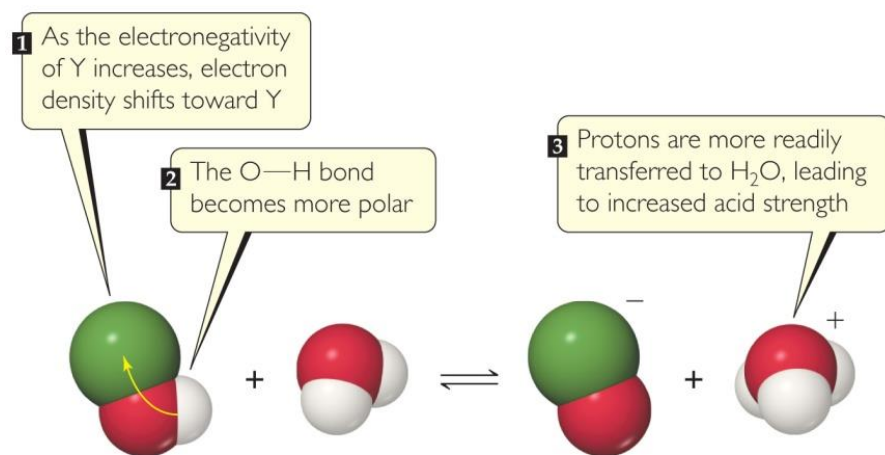
| 4A                                                 | 5A                                                                   | 6A                                                                | 7A                                                   |                               |
|----------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------|-------------------------------|
| <b>CH<sub>4</sub></b><br>Neither acid<br>nor base  | <b>NH<sub>3</sub></b><br>Weak base<br>$K_b = 1.8 \times 10^{-5}$     | <b>H<sub>2</sub>O</b>                                             | <b>HF</b><br>Weak acid<br>$K_a = 6.8 \times 10^{-4}$ | Increasing acid strength<br>↓ |
| <b>SiH<sub>4</sub></b><br>Neither acid<br>nor base | <b>PH<sub>3</sub></b><br>Very weak base<br>$K_b = 4 \times 10^{-28}$ | <b>H<sub>2</sub>S</b><br>Weak acid<br>$K_a = 9.5 \times 10^{-8}$  | <b>HCl</b><br>Strong acid                            |                               |
|                                                    |                                                                      | <b>H<sub>2</sub>Se</b><br>Weak acid<br>$K_a = 1.3 \times 10^{-4}$ | <b>HBr</b><br>Strong acid                            |                               |
| Increasing acid strength<br>→                      |                                                                      |                                                                   |                                                      |                               |

# Binary Acids

- ◆ As the element A increases in size → The strength of an H – A bond tends to decrease → The acidity **increases down a group**
  - HCl is a stronger acid than HF
  - H<sub>2</sub>S is a stronger acid than H<sub>2</sub>O
- ◆ Bond polarity is the major factor determining acidity: acidity increases as the electronegativity of the element A increases
  - C – H bond is essentially nonpolar, CH<sub>4</sub> shows no tendency to form H<sup>+</sup> and CH<sub>3</sub><sup>-</sup> ions.
  - N – H bond is polar, but NH<sub>3</sub> has a lone pair of electrons on the nitrogen atom that dominates its chemistry, so NH<sub>3</sub> acts as a base rather than an acid

# Oxyacids

In oxyacids, in which an –OH is bonded to another atom, Y, the more electronegative Y is, the more acidic the acid.



| Substance         | Y—OH  | Electronegativity<br>of Y | Dissociation<br>constant    |
|-------------------|-------|---------------------------|-----------------------------|
| Hypochlorous acid | Cl—OH | 3.0                       | $K_a = 3.0 \times 10^{-8}$  |
| Hypobromous acid  | Br—OH | 2.8                       | $K_a = 2.5 \times 10^{-9}$  |
| Hypoiodous acid   | I—OH  | 2.5                       | $K_a = 2.3 \times 10^{-11}$ |
| Water             | H—OH  | 2.1                       | $K_w = 1.0 \times 10^{-14}$ |

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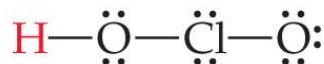
# Oxyacids

Hypochlorous



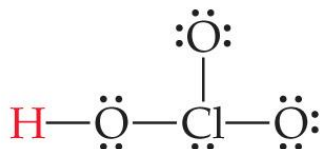
$$K_a = 3.0 \times 10^{-8}$$

Chlorous



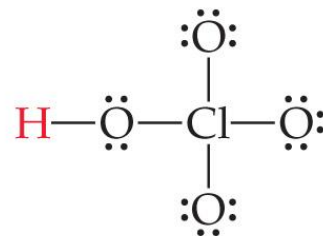
$$K_a = 1.1 \times 10^{-2}$$

Chloric



Strong acid

Perchloric



Strong acid

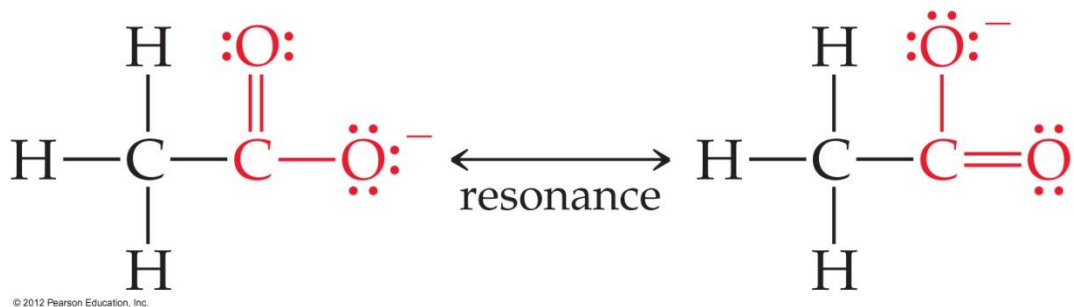
Increasing acid strength



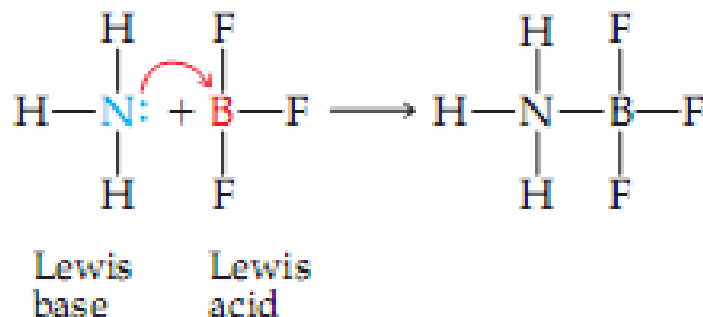
For a series of oxyacids, acidity increases with the number of oxygens.

# Carboxylic Acids

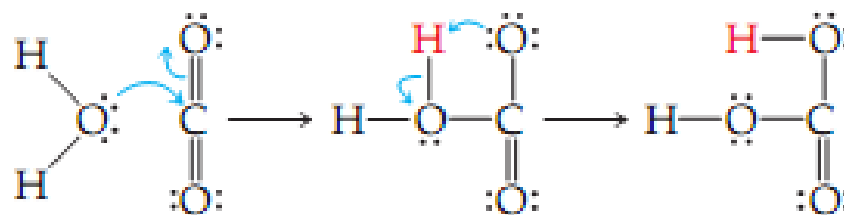
Resonance in the conjugate bases of carboxylic acids stabilizes the base and makes the conjugate acid more acidic.



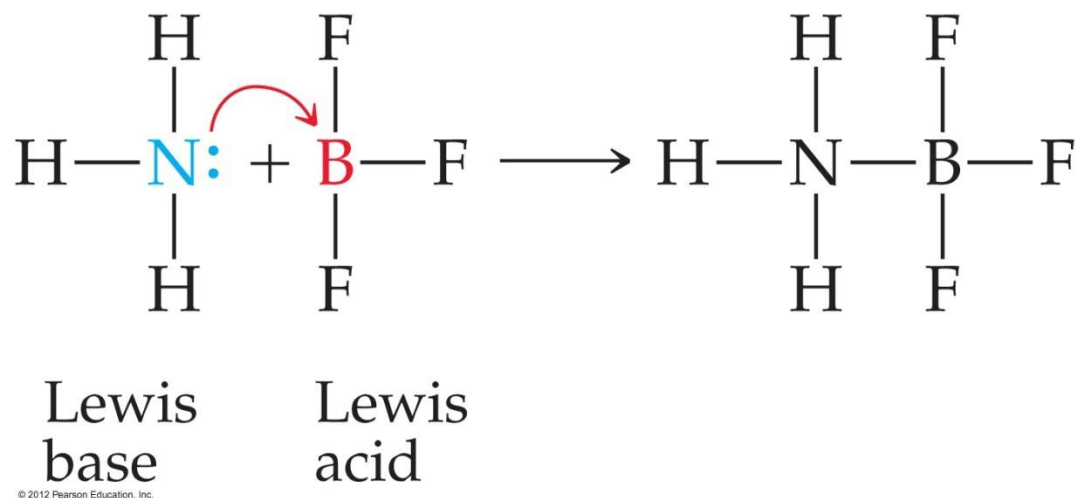
## 16.11 | Lewis Acids and Bases



- ◆ **Lewis acids** are defined as electron-pair acceptors.
- ◆ Atoms with an empty valence orbital can be Lewis acids.



# Lewis Bases



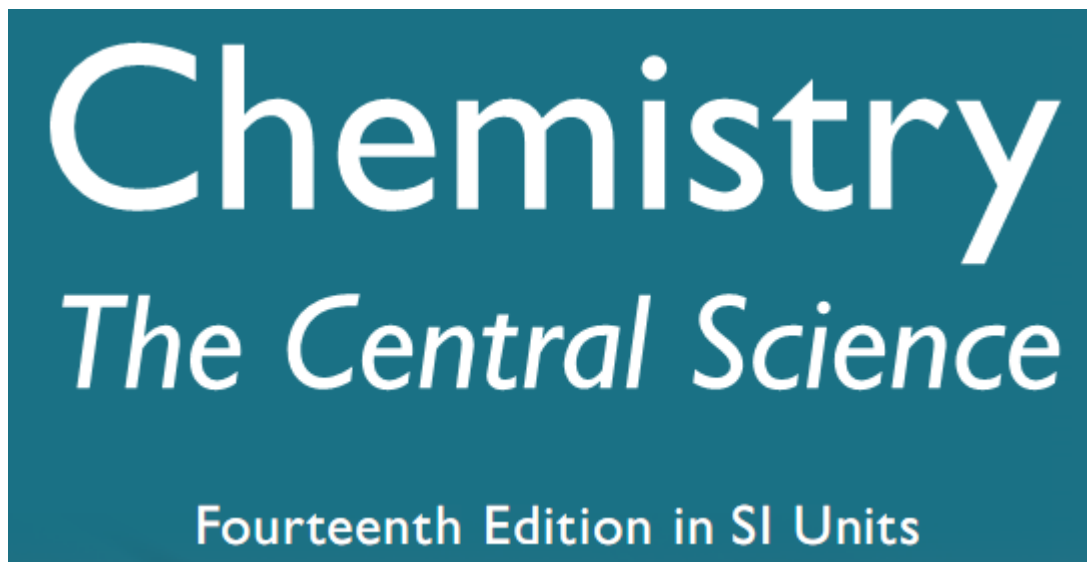
- ◆ **Lewis bases** are defined as electron-pair donors.
- ◆ Anything that could be a Brønsted–Lowry base is a Lewis base.
- ◆ Lewis bases can interact with things other than protons, however.

# Homeworks

**16.119**

**16.126**

***Due: August 14<sup>th</sup> 2019***



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***It's a quick reminder that Group 1 and Group 2 will present their research topics on Wednesday August 14<sup>th</sup> 2019***